



Econometrics applied to finance

Time series, GARCH, non-linearities: the tools used to model financial data.

GARCH(1,1) – MODELING VOLATILITY

$$\sigma_t^2 = \omega + \alpha \cdot \varepsilon_{t-1}^2 + \beta \cdot \sigma_{t-1}^2$$

TIME SERIES

- Stationarity: stable mean and variance over time, a key condition for modeling.
- ACF/PACF: tools to identify a series' autocorrelation structure.
- A price series is rarely stationary; its returns often are much more so.

VOLATILITY ECONOMETRICS

- ARCH/GARCH: model volatility clustering (calm and turbulent periods that group together).
- σ depends on its own past and on recent shocks.
- The backbone of most practical risk and options pricing models.

NON-LINEAR MODELS

- Regime-switching models (Markov-switching): dynamics change depending on the market state.
- Threshold models: behavior shifts once a critical threshold is crossed.
- Capture dynamics that classic linear models miss.

UNIT ROOT TESTS

- ADF test (Augmented Dickey-Fuller): checks whether a series is stationary.
- A must before any regression on time series (otherwise spurious regression).
- Complemented by cointegration tests for long-run relationships between series.

Who uses what?

QUANT RESEARCHER

Models price and volatility dynamics for pricing and risk purposes.

RISK MODELER

Uses GARCH to forecast short-horizon volatility within VaR models.

ECONOMETRICIAN

Tests stationarity and cointegration before any macro modeling.

DATA SCIENTIST

Combines classic econometrics with machine learning for series forecasting.

KEY TAKEAWAYS

1. Ignoring non-stationarity is mistake #1 in financial econometrics — it leads to spurious regressions with misleading results.
2. Volatility clustering (calm/turbulent periods grouped together) is the whole reason GARCH models exist.
3. Non-linear models don't replace linear ones: they complement them when the relationship between variables changes depending on the market context.



Comparison table

MODEL	OBJECT MODELED	TYPICAL USE
AR / MA / ARMA	Level of a stationary series	Short-term return forecasting
ARCH / GARCH	Conditional variance	Volatility forecasting, VaR
Markov-switching	Regime change	Identifying market phases
VAR (vector autoregression)	Several interdependent series	Macro analysis, shock transmission

Real-world examples

VOL FORECASTING WITH GARCH

An options desk calibrates a GARCH(1,1) on returns to anticipate 1-day volatility and adjust its pricing.

PAIRS TRADING

Two cointegrated stocks justify a pairs trading strategy: betting on the mean reversion of their price spread.

MACRO VAR

A VAR model links GDP, inflation and the policy rate to simulate the effect of a monetary policy shock.

REGIME CHANGE

A Markov-switching model automatically detects the shift from a low- to a high-volatility regime in equity markets.

Quick glossary

STATIONARITY

Property of a series whose statistical characteristics don't change over time.

AUTOCORRELATION

Correlation of a series with its own past values.

COINTEGRATION

Stable long-run relationship between two series that are individually non-stationary.

HETEROSCEDASTICITY

Residual variance that changes over time, typical of financial series.

UNIT ROOT

Characteristic of a non-stationary series, tested via ADF.

BACKTESTING

Validating a model on historical data before production use.

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